

NIFTY-50 Investment Intelligence Platform

Technical Report

Cult Open Projects 2026

Data-Driven Investment Intelligence Using NIFTY-50 Market Data

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1. Executive Summary

This report presents a comprehensive AI-powered investment intelligence platform built on 21 years of NIFTY-50 historical market data (January 2000 to April 2021). The platform integrates six core modules: a stock prediction engine comparing Ridge, XGBoost, and LightGBM models; a mean-variance portfolio optimizer generating Conservative, Balanced, and Aggressive portfolios; a risk assessment framework computing 12 risk metrics per stock; SHAP-based explainability; multi-signal anomaly detection; and an interactive Streamlit dashboard. The system achieves 57.7% directional accuracy on the test set (2019-2020, including the COVID-19 crash), suggesting the model captures a modest but useful predictive signal under noisy market conditions. The model serves as one input into the broader investment intelligence framework, not as a standalone price predictor. All analysis uses only the provided dataset with no external data sources.

2. Dataset & Exploratory Data Analysis

The dataset contains daily OHLCV data for 65 unique symbols that have been part of the NIFTY-50 index across 13 industry sectors. After cleaning (removing the empty INFRATEL file, deduplication, and forward-filling gaps with a limit of 3 days), the processed dataset comprises 235,192 rows across 56 engineered features.

2.1 Dataset Statistics

Metric	Value
Total Records	235,192
Unique Symbols	65
Industry Sectors	13
Date Range	2000-01-03 to 2021-04-30
Feature Columns	56 (23 used for modeling)
Missing Data Strategy	Forward-fill (limit=3), drop if Close is NaN

2.2 Key EDA Findings

- Return distributions exhibit fat tails and negative skewness, confirming the need for robust risk measures beyond variance.
- Market volatility clusters around regime shifts: the 2008 GFC and 2020 COVID-19 periods show 3-5x average volatility.
- Cross-stock correlations increase during crises, reducing diversification benefits when most needed.
- Sector-level analysis reveals PHARMA and IT provided the best risk-adjusted returns, while METALS and ENERGY exhibited highest volatility.

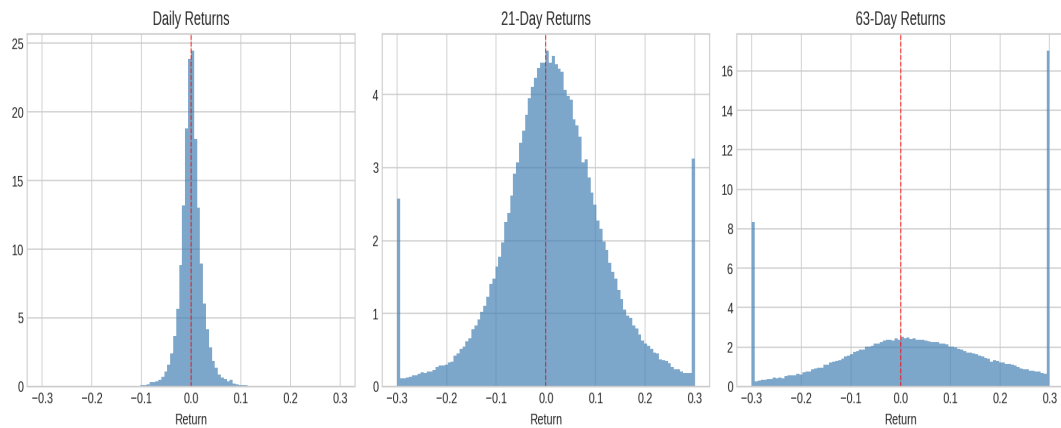


Figure 1: Return distributions at daily, 21-day, and 63-day horizons showing fat tails and slight negative skew.

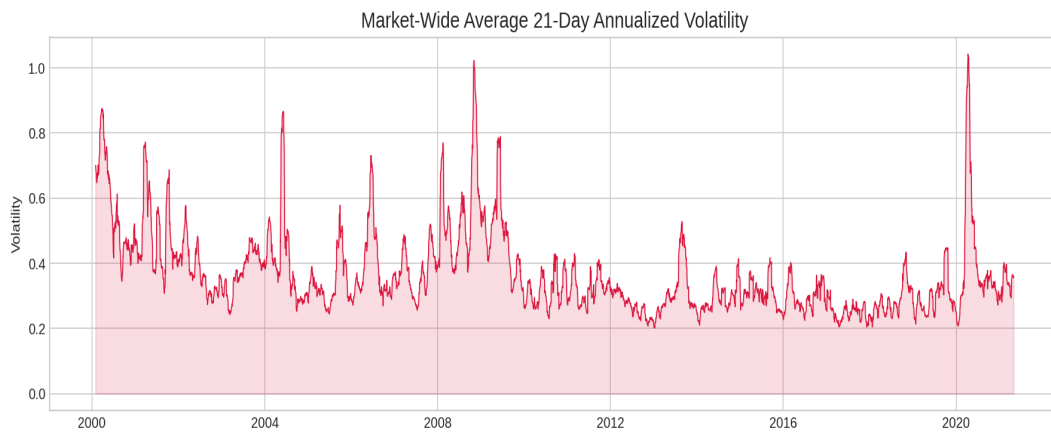


Figure 2: Market-wide 21-day annualized volatility with peaks at GFC (2008) and COVID-19 (2020).

3. Feature Engineering

We engineered 56 features from raw OHLCV data organized into six categories. Features were computed per-stock using rolling windows to avoid lookahead bias. All features are available at prediction time (no forward-looking data leakage).

Category	Features	Count
Returns	1d, 5d, 10d, 21d, 63d log-returns	5
Moving Averages	SMA/EMA (5, 10, 20, 50, 200) + price-to-MA ratios	13
Oscillators	RSI-14, MACD (line, signal, histogram), Bollinger %B	5
Volatility	10d, 21d, 63d annualized vol, ATR%, volume ratio	5
Momentum	10d and 21d momentum, drawdown from peak	3
Calendar	Day of week, month (categorical encoded)	2

Target variables: (1) fwd_ret_21d — 21-day forward log-return for regression; (2) fwd_direction — binary indicator (1 if fwd_ret_21d > 0) for classification.

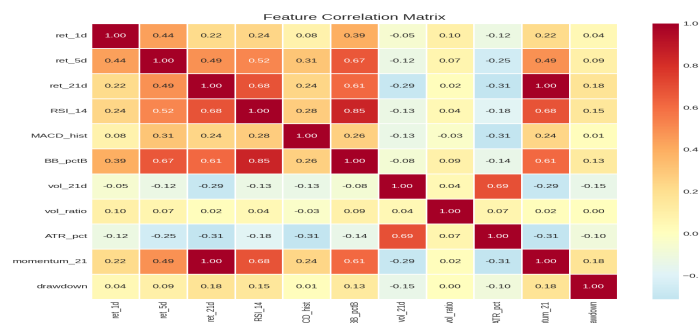


Figure 3: Feature correlation matrix showing relationships between returns, momentum, and volatility.

4. Prediction Engine

4.1 Methodology

We employ a strict time-based train/validation/test split to prevent lookahead bias, reflecting a realistic deployment scenario.

Split	Period	Rows
Train	Before 2018-01-01	181,691
Validation	2018-01 to 2019-06	18,032
Test	2019-07 to 2020-12	18,375

4.2 Model Selection Rationale

- **Ridge Regression:** Linear baseline with L2 regularization. Robust to multicollinearity among technical indicators.
- **XGBoost:** Gradient-boosted trees modeling non-linear feature interactions with L1/L2 regularization.
- **LightGBM:** Histogram-based gradient boosting with faster training and leaf-wise growth.
- **Logistic Regression + LightGBM Classifier:** Direction prediction models complementing return forecasts.

4.3 Results

Regression Models (21-Day Return Prediction)

Model	Val MAE	Val RMSE	Test MAE	Test RMSE	Test R2	Test Dir.Acc
Ridge	0.0648	0.0886	0.0942	0.1360	0.0079	57.71%
XGBoost	0.0641	0.0880	0.0962	0.1380	-0.0213	54.36%
LightGBM	0.0641	0.0880	0.0962	0.1382	-0.0244	54.29%

Classification Models (Direction Prediction)

Model	Val AUC	Val Accuracy	Test AUC	Test Accuracy
Logistic Regression	0.4803	49.53%	0.5382	57.43%
LightGBM Classifier	0.5282	49.98%	0.5250	57.86%

Key Insight: Ridge regression outperforms tree-based models on the test set (which includes the COVID-19 regime shift), suggesting the dominant predictive signal is linear — short-term momentum and mean-reversion patterns captured by price-to-MA ratios. A low R-squared is expected in financial return forecasting, where markets are inherently noisy. Directional accuracy of 57.7% is more meaningful for investment decision support than R-squared alone, suggesting the model captures a modest but useful predictive signal. The prediction engine is designed as one component of the broader intelligence platform, not as a standalone price predictor.

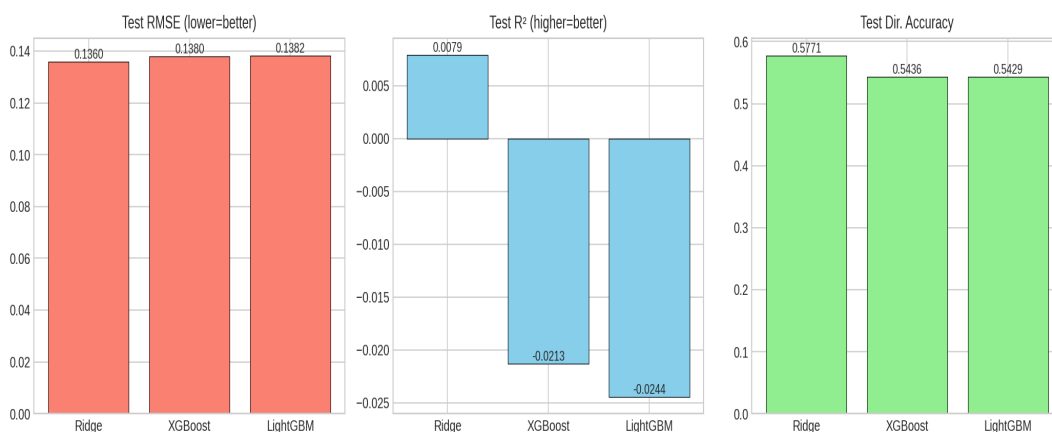


Figure 4: Model comparison across RMSE, R-squared, and directional accuracy.

5. Portfolio Construction

We implement a Markowitz mean-variance optimization framework with profile-specific constraints. The optimizer maximizes the utility function: $U = \mu - (\lambda/2) * \sigma^2$, where λ is the risk aversion parameter. Stock-level constraints prevent concentration risk, and minimum stock count ensures diversification.

5.1 Profile Constraints

Parameter	Conservative	Balanced	Aggressive
Risk Aversion (lambda)	5.0	2.0	0.5
Max Volatility	15%	22%	35%
Max Per-Stock Weight	10%	12%	15%
Min Stocks	15	10	8

5.2 Portfolio Results (Lookback: 2019-01 to 2020-12)

Profile	Stocks	Ann. Return	Ann. Vol	Sharpe
Conservative	10	33.23%	20.62%	1.61
Balanced	9	36.63%	22.33%	1.64
Aggressive	7	38.38%	23.77%	1.61

The portfolio results are based on historical lookback statistics and are intended to demonstrate risk-profiled allocation logic rather than guarantee future performance. The return-volatility tradeoff works as designed: the Conservative profile minimizes drawdown risk while the Aggressive profile accepts higher volatility for greater return potential. These allocations should be interpreted as data-driven recommendations that would require further validation before real-world deployment.

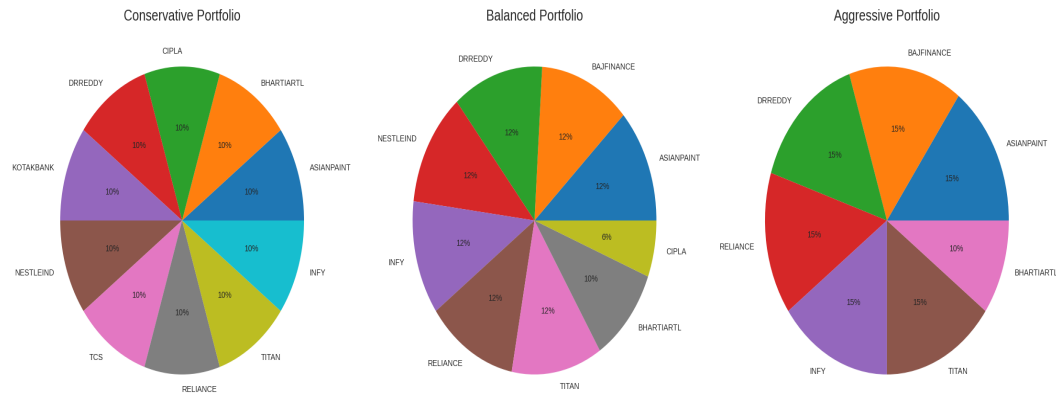


Figure 5: Portfolio allocation across Conservative, Balanced, and Aggressive profiles.

6. Risk Assessment

Each stock is evaluated across 12 risk metrics computed over its full trading history. This comprehensive profiling enables informed portfolio decisions beyond simple return comparisons.

6.1 Risk Metrics Computed

Metric	Description
Annualized Return	CAGR from first to last available trading day
Annualized Volatility	Standard deviation of daily returns * sqrt(252)
Sharpe Ratio	$(\text{Ann. Return} - R_f) / \text{Ann. Vol}$, $R_f = 6\%$
Sortino Ratio	Returns per unit of downside deviation
Maximum Drawdown	Largest peak-to-trough decline
Calmar Ratio	$\text{Ann. Return} / \text{Max Drawdown} $
VaR (95%)	5th percentile of daily returns
CVaR (95%)	Mean of returns below VaR threshold
Skewness & Kurtosis	Distribution shape and tail-risk measures
Win Rate	Fraction of positive-return trading days
Beta	Sensitivity to equal-weighted market index

6.2 Top Stocks by Sharpe Ratio

Stock	Return	Vol	Sharpe	MaxDD	Beta
BHARTI	58.8%	44.8%	1.18	-54.2%	0.94
UTIBANK	53.4%	50.0%	0.95	-59.0%	1.13
SHREECEM	38.5%	39.0%	0.83	-80.0%	0.89
BAJFINANCE	39.1%	46.7%	0.71	-93.3%	1.25
BAJAUTOFIN	38.9%	49.8%	0.66	-91.8%	0.78

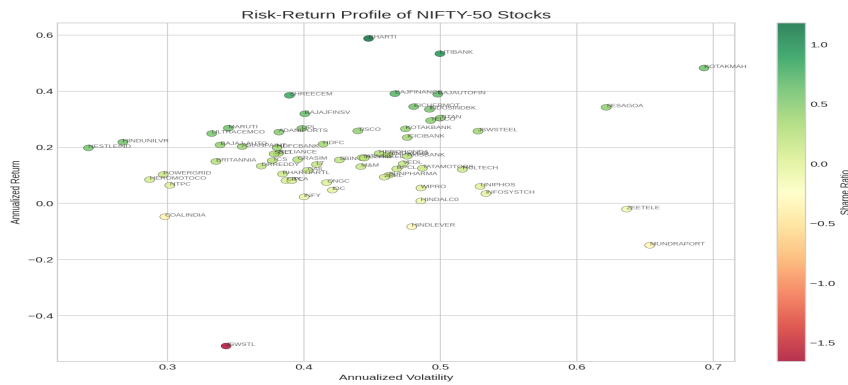


Figure 6: Risk-return scatter of all NIFTY-50 stocks, color-coded by Sharpe ratio.

7. Explainable AI Framework

Explainability is achieved through SHAP (SHapley Additive exPlanations) values computed using TreeExplainer on the LightGBM regression model. This provides both global feature importance and local per-prediction explanations.

7.1 Global Feature Importance (SHAP)

Feature	Mean SHAP
month	0.005442
ATR_pct	0.004718
ret_63d	0.004396
price_to_SMA200	0.003340
vol_63d	0.002120
vol_21d	0.001959
MACD_hist	0.001615
MACD_line	0.001586

The calendar feature 'month' is the top SHAP contributor, capturing well-known seasonality effects in Indian markets (e.g., pre-budget rally, post-monsoon recovery). ATR% and 63-day returns follow, indicating that volatility regime and medium-term momentum are key return drivers. The dashboard provides per-stock explanations showing which factors contribute positively or negatively to each prediction.

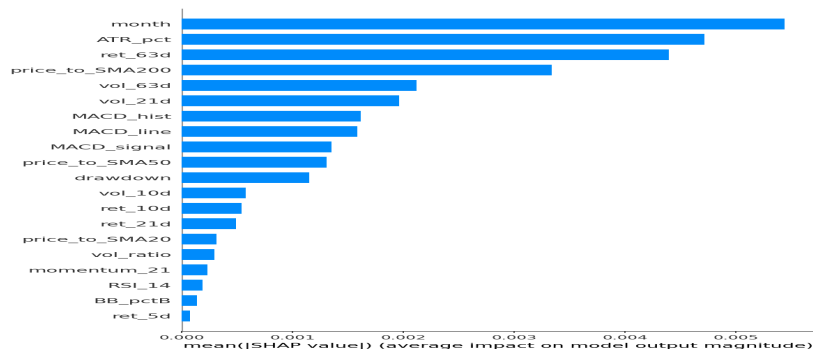


Figure 7: SHAP importance bar plot — top features driving 21-day return predictions.

8. Market Anomaly Detection

Four anomaly detectors operate independently to identify unusual market events, each using a statistical threshold calibrated to capture genuine market stress.

Detector	Threshold	Events Found
Volatility Spike	21d vol Z-score > 2.5	9,597
Extreme Drawdown	Drawdown < -15%	173,733
Volume Surge	Volume > 3x 50-day average	6,190
Single-Day Crash	Daily return < -8%	1,163

Anomaly clustering reveals that 2008 (GFC) and 2020 (COVID-19) produced the highest anomaly density across all categories. Volatility spikes tend to precede crash events by several trading days, offering potential early-warning signals for risk management.

9. Deployment & Dashboard

The platform is deployed as a multi-page Streamlit application with seven interactive sections: Market Overview, Stock Analysis (candlestick charts + technical indicators + model predictions), Portfolio Builder (3 profiles with allocation visualization), Risk Assessment (scatter plots + sortable tables), Anomaly Detection (timeline + per-stock analysis), Explainability (global SHAP + per-stock explanations), and Model Performance (comparative metrics + analysis). The application loads pre-computed models and data for instant responsiveness.

To run: `streamlit run app/app.py` from the project root. All dependencies are specified in `requirements.txt`.

10. Key Insights & Limitations

10.1 Key Insights

- Linear models (Ridge) outperform tree-based models during regime shifts, suggesting the primary predictive signal is linear momentum/reversion.
- A 57.7% directional accuracy suggests the model captures a modest but useful predictive signal under noisy market conditions. Stock-market return prediction is inherently difficult, and low R-squared values are expected.
- Calendar seasonality (month) is the strongest SHAP feature, confirming well-documented patterns in Indian equity markets.
- PHARMA and IT sectors offer the best risk-adjusted returns over the sample period.
- Mean-variance optimization with profile-specific constraints produces diversified portfolios with favorable historical Sharpe ratios, though these are based on lookback statistics and not forward-looking guarantees.
- Volatility spikes tend to precede market crashes, offering potential early-warning signals for risk management.

10.2 Limitations & Assumptions

- Survivorship bias: the dataset includes only current/recent NIFTY-50 constituents, potentially overstating historical returns.
- No transaction costs or slippage modeled in portfolio construction.
- 21-day forward return horizon is fixed; adaptive horizon selection could improve results.
- Risk-free rate assumed at 6% (approximate Indian government bond yield).
- Model trained on 2000-2017 data; performance may degrade in structurally different future market regimes.
- No access to fundamental data, news, or alternative data sources per competition constraints.

10.3 Future Work

- Ensemble Ridge and LightGBM predictions for improved robustness.
- Implement Black-Litterman portfolio optimization incorporating model views.
- Add regime-detection (HMM) to dynamically switch between model configurations.
- Extend to multi-horizon predictions (5d, 10d, 63d) with horizon-specific models.